
Software Requirements Specification

for

IntelliRide: Heuristic-Based Ride-Sharing & Carpool Matching System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
IntelliRide Team	13th February, 2026	Initial draft of SRS document	V1.0

1. Introduction

University students struggle with urban transportation due to the rising fuel costs, traffic congestion, limited parking, and environmental concerns. While other commercially available platforms for road traversal, such as Uber and Careem, provide general transportation services, they are not specifically designed for structured university environments where students have to follow fixed schedules and routines.

Existing systems tend to focus on vague ML models, causing users to have difficulty understanding and trusting said systems. When it comes to students, they can hesitate to use these systems as it is possible that they do not understand how the matches are made or whether the safety and schedule compatibility are properly considered.

IntelliRide addresses these gaps by introducing a transparent and heuristic-based ride-sharing approach that is tailored for university students, focusing on schedule compatibility, route similarity, trust scores, and time constraints.

1.1 Purpose

This Software Requirements Specification (SRS) document defines the requirements for the IntelliRide system, a university-focused heuristic-based ride-sharing & carpool matching system. The software described here is Version 1.0 (Initial Release).

The primary purpose of IntelliRide is to provide an intelligent and transparent ride matching system for university students that would reduce the transportation costs and improve the trust and safety of the system. This is done by developing a schedule-aware heuristic algorithm to optimize the matching results.

This SRS covers the entire system including:

- User registration and authentication
- Driver trip creation
- Passenger ride requests
- Heuristic-based matching algorithm
- Trust and rating system
- Administrative controls

1.2 Document Conventions

This Software Requirements Specification (SRS) follows the IEEE 830 standard format for documenting requirements.

Headings & Numbering: Sections are structured hierarchically (e.g., 1, 1.1, 1.2) for clarity.

Fonts & Styles:

- Bold is used for section headings and key terms.
- Italics highlight examples or notes.
- Monospace font is used for code snippets, database schemas, and JSON structures.

Terminology: Terms like Driver, Passenger, and Admin are capitalized consistently to indicate defined entities within the system.

References: Citations are formatted using APA style.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following stakeholders:

Drivers & Passengers: Primary users of the system.

University Stakeholders: To evaluate system suitability in a university setting.

Development Team: Understanding the system behavior/constraints and to be responsible for the design and implementation of the system.

System Administrators: Responsible for deployment and maintenance of the system.

Evaluators/Instructors: For academic project review and to derive test cases.

Reading suggestions:

Users interested in system behavior should focus on Section 2 (Functional Requirements).

Developers and testers should read Sections 2, 3, and 4.

Stakeholders reviewing scope can refer to Section 1.4 (Product Scope).

References and supporting documents are listed in Section 1.5.

1.4 Product Scope

The proposed system is a university-focused ride-sharing system that matches student drivers and passengers based on several key features:

Drivers can create trips with origin, destination, departure time, and seat availability.

Passengers can request rides based on student schedule and location.

The system generates ranked matches using a transparent scoring formula.

Users can view explanations displayed about the match scores.

A trust and feedback system enhances safety.

Business Benefits:

Reduce student transportation costs.

Decreases parking demand and traffic congestion.

Encourages environmentally sustainable commuting.

Increases transparency compared to traditional ride-sharing systems.

Constraints & Scope Boundaries (out of scope):

Real-time GPS tracking.

In-app payments.

Integration with external mapping APIs.

AI-based behavioral prediction.

1.5 References

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2. Functional Requirements

Driver Requirements

REQ-1 (Driver Registration): Drivers shall register by using a verified university email address.

Error: An error message will be displayed if the entered email address is Invalid or a duplicate.

REQ-2 (Trip Creation): Drivers shall create a trip specifying origin, destination, departure time, and available seats.

Error: Rejection of request shall occur in case of invalid time or empty fields.

REQ-3 (Trip Modification): Drivers shall edit or cancel trips before departure.

REQ-4 (Ride Approval): Drivers shall approve or reject passenger requests.

REQ-5 (Seat Management): The system shall prevent bookings exceeding the available seats.

Passenger Requirements

REQ-6 (Passenger Registration): Passengers shall register using a verified university email address.

REQ-7 (Ride Request): Passengers shall submit ride requests specifying their desired time and location.

Error: Rejection of request shall occur in case of overlapping ride requests.

REQ-8 (View Ranked Matches): Passengers shall view ranked driver matches with score explanations.

REQ-9 (Ride Confirmation): Passengers shall receive a confirmation message once a driver has approved a request.

Matching Engine Requirements

REQ-10 (Match Score calculation): The system shall calculate a match score based on the predefined heuristic criteria.

REQ-11 (Score Transparency): The system shall display a breakdown of scoring components.

REQ-12 (Ranking): The system shall rank drivers in a descending order of match scores.

REQ-13 (Threshold Filtering): Matches exceeding maximum threshold time shall be excluded.

Trust & Feedback Requirements

REQ-14 (User Rating): Users shall be able to rate each other after ride completion.

REQ-15 (Trust Score Update): The system shall update trust scores dynamically.

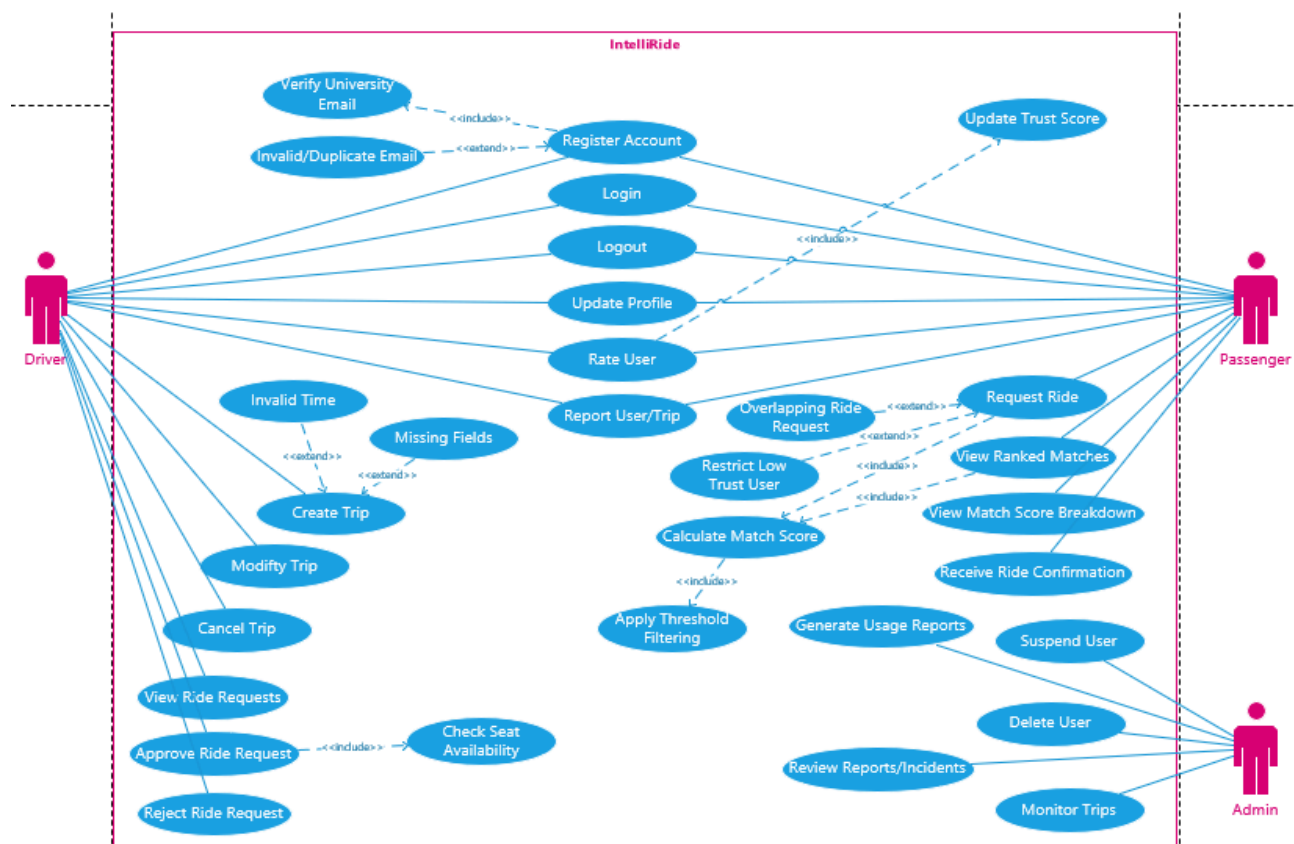
REQ-16 (Low Trust Restriction): Users below the minimum trust threshold shall be restricted.

Admin Requirements

REQ-17 (User Management): Admin shall be able to suspend or delete accounts.

REQ-18 (Trip Monitoring): Admin shall monitor all trips and ride activities.

REQ-19 (Report Generation): Admin shall generate usage statistics reports.



3. Non-Functional Requirements

3.1 Performance Requirements

NFR-1: The system shall display ranked matches for a ride request within 5 seconds under normal load.

NFR-2: The system shall be accessible at least between 6:00 AM and 10:00 PM during academic days.

NFR-3: The system shall support at least 100 concurrent users performing searches/requests without exceeding the 5 second response time requirement.

NFR-4: The system shall load core pages (Login, Dashboard, Trip Creation, Ride Request) in ≤ 3 seconds on a standard university network connection.

NFR-5: The system shall complete match score computation for a single ride request within 3 seconds for up to 200 candidate trips.

3.2 Safety Requirements

NFR-6: The system shall restrict access to ride creation and ride requesting features to verified university users only.

NFR-7: The system shall require all users to acknowledge and accept a liability disclaimer during registration, stating that IntelliRide acts solely as a ride matching platform, and holds no liability for any accidents, injuries, property damage, misconduct, or any form of disputes that may emerge during the ride sharing activity.

NFR-8: The system shall provide a “Report User/Trip” option during an active booking and after ride completion.

NFR-9: The system shall automatically restrict users whose trust score falls below the minimum threshold defined by the system policy (check Business rules), preventing them from requesting/approving rides.

NFR-10: The system shall log ride approvals, cancellations, and reports to support admin review and incident tracking

3.3 Security Requirements

NFR-11: The system shall require users to register/login using a verified university email (@aus.edu) and prevent duplicate accounts.

NFR-12: Passwords shall be stored using a secure one way hashing algorithm.

NFR-13: The system shall enforce role based access control (Driver/Passenger/Admin) such that only the admin can suspend/delete accounts and generate reports.

NFR-14: Sensitive data (credentials, user identifiers, reports) shall not be stored or transmitted in plain text.

NFR-15: The system shall record audit logs for critical actions (login attempts, trip creation/modifications/cancellations, ride approvals/rejections, and account suspensions).

3.4 Software Quality Attributes

NFR-16: The UI shall be easy to learn such that a first time user can request or create a trip within 10 minutes without external guidance (including registration process...).

NFR-17: The system shall present match results with a clear breakdown of the score and a concise explanation.

NFR-18: The system shall maintain at least 95% uptime during an academic semester, excluding scheduled maintenance.

NFR-19: The system shall handle unexpected failures without data corruption (for example, an incomplete trip creation shall not create in the database an incomplete trip data entry).

NFR-20: The software shall be modular such that the heuristic scoring can be updated without redesigning unrelated modules such as the user authentication UI.

NFR-21: The system shall run on majorly used web browsers (Chrome, Safari, Edge) and maintain functional layout on common mobile resolutions.

NFR-22: The system shall allow scaling of stored users/trips to at least 5000 users and a minimum of users*5 trips per semester without changing core architecture.

NFR-23: The system shall comply with all applicable UAE RTA regulations governing private vehicle usage and ride sharing activities.

3.5 Business Rules

NFR-24: Only users with verified university email addresses shall be allowed to create an account.

NFR-25: A driver may not approve ride requests that exceed the vehicle's available seats (enforced constraint).

NFR-26: Trips may only be modified/canceled by the driver before the departure time.

NFR-27: Admin has the authority to suspend/delete accounts, monitor trips, and generate reports, while regular users can't access administrative reports.

NFR-28: Users with trust score below the minimum threshold (unconfirmed example: 2.5/5) shall be restricted from requesting/approving rides until reviewed by admin.

4. Domain Requirements

DR-1: Database Requirements

The system database shall:

- Store students' schedules on a fixed per semester basis.
- Store user profiles including role (Driver/Passenger), trust score, and ride history.
- Store trip data including origin, destination, depart time, seat availability.
- Store ride request status (Pending, approved, rejected, completed).
- Maintain referential integrity between users, ride requests, and ratings.

DR-2: Ride Time Constraints

Ride matching shall respect university schedule constraints such that:

- Matching must consider class start times.
- Time deviation between driver and passenger shall not exceed a predefined maximum threshold (ex. 15 mins).

DR-3: Seat Allocation Constraint

For each trip, Passenger count in a ride request can not exceed the number of seats available in a singular ride.

DR-4: Legal and Regulatory Constraints

The system shall:

- Comply with applicable UAE RTA regulations regarding private vehicle ride sharing.
- Operate as a cost sharing platform and not as a commercial ride hailing service.
- Respect university transportation and student conduct policies.